

SPRINT BACKLOG & PRODUCT BACKLOG

Enterprise Manufacturing Data Platform (EMDP)

1. Product Backlog

Epic	Description	Planned Sprint
Project Initiation	Define business problem, stakeholders, requirements, scope, architecture, repository, and project governance.	Sprint 0
Data Ingestion	Ingest heterogeneous source data into PostgreSQL Bronze/staging using Pentaho.	Sprint 1
Data Profiling & Warehousing	Profile, cleanse, validate, transform, and model data into Silver and Gold warehouse layers.	Sprint 2
Governance & Analytics	Implement lineage, metadata, orchestration, validation, and business dashboards.	Sprint 3

2. Sprint 0 – Project Initiation & Architecture

Business Objective: Understand the manufacturing ecosystem and prepare the project plan and architecture before development begins.

ID	Task	Owner	Status
S0-1	Study the business problem	Data Engineer	Done
S0-2	Identify stakeholders	Data Engineer	Done
S0-3	Prepare Business Requirement Document (BRD)	Data Engineer	Done
S0-4	Identify available data sources	Data Engineer	Done
S0-5	Define project scope	Data Engineer	Done
S0-6	Design high-level architecture	Data Engineer	Done
S0-7	Create Git repository	Data Engineer	Done
S0-8	Create project folder structure	Data Engineer	Done
S0-9	Prepare Product Backlog	Data Engineer	Done

3. Sprint 0 Deliverables

- Business Requirement Document (BRD)
- Solution Architecture
- Git Repository
- Sprint Backlog
- Project Charter

4. Sprint 1 – Data Discovery & Ingestion

Business Objective: Build the Raw Data Landing Zone by integrating data from multiple enterprise systems.

ID	Task	Owner	Status
S1-1	Analyze all source systems	Data Engineer	To Do
S1-2	Prepare Data Dictionary	Data Engineer	To Do
S1-3	Develop Pentaho transformations to ingest CSV, Excel, JSON, XML, and SQL data	Data Engineer	To Do
S1-4	Load raw data into PostgreSQL staging tables	Data Engineer	To Do
S1-5	Implement basic logging and exception handling	Data Engineer	To Do
S1-6	Commit code to Git after successful ingestion	Data Engineer	To Do

5. Sprint 1 Deliverables

- Source Inventory
- Data Dictionary
- Pentaho ETL Pipelines
- Staging Database
- Git Commit History

6. Sprint 1 Acceptance / Definition of Done

- Business requirements addressed for the sprint.
- Pentaho ETL pipelines execute successfully.
- Data quality checks are documented.

- PostgreSQL staging tables are populated correctly.
- Source-to-target mappings are updated.
- Code is committed to Git with meaningful commit messages.
- Sprint documentation is completed.
- Sprint review presentation is delivered.

7. Working Rule for Sprint Execution

Sprint tasks will be marked Done only after the corresponding work has been implemented, tested, documented, and verified. Until then, tasks remain To Do or In Progress. This ensures the backlog reflects the actual project state.